

HYDROGENATED NBR (HNBR) SYNTHESIS, MOLECULAR-WEIGHT CONTROL, AND CATALYST-REMOVAL PATENT LANDSCAPE

1. ABSTRACT

This report synthesizes a set of patents relevant to hydrogenated nitrile rubber (HNBR), with emphasis on how NBR precursors are converted, modified, or post-treated to improve processability, residual unsaturation, molecular weight, and catalyst cleanliness. The target material is hydrogenated NBR, including partially and highly hydrogenated grades, where the key performance drivers are saturation level, thermal stability, oil resistance, and manageable Mooney viscosity. Across the patent set, two major technical themes emerge. First, several patents address direct hydrogenation of NBR in solution using homogeneous or heterogeneous transition-metal catalysts, with explicit attention to hydrogen pressure, temperature, catalyst composition, and co-catalyst pretreatment to control hydrogenation rate and suppress gelation or excessive molecular-weight drift. Second, other patents use metathesis or segment-interchange chemistry to reduce molecular weight before or during hydrogenation, thereby producing lower-viscosity HNBR with narrower molecular-weight distributions and improved processability. A third theme is downstream purification: one patent focuses on removing residual hydrogenation catalyst from HNBR cements by chelation, semi-coagulation, and washing, while another addresses phosphine/phosphine oxide and halogen residues in HNBR made with phosphine-containing hydrogenation catalysts. The landscape also includes application-driven disclosures where HNBR is selected for swellable packers, transmission belts, and other elastomeric systems, reinforcing the importance of controlled hydrogenation degree and low residual impurities. Overall, the evidence indicates that the most effective HNBR synthesis strategies combine controlled catalyst chemistry, moderate-to-high hydrogenation pressure, elevated solution-phase temperatures, and post-hydrogenation purification or pre-hydrogenation molecular-weight reduction to achieve a balance of saturation, low Mooney viscosity, and robust end-use performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route in which an unsaturated nitrile rubber is first subjected to olefin metathesis with a ruthenium- or osmium-based catalyst system and then hydrogenated to form low-molecular-weight HNBR. The metathesis step is used to reduce molecular weight and narrow the molecular-weight distribution before hydrogenation, improving processability while retaining the HNBR microstructure. - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 15 wt.-% solid — Example 2 - Cement concentration: 6 wt.-% solid — Example 3 - Cement concentration: 6 wt.-% solid — Example 4 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - 1-Hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 / Examples 1-4 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - MWD: 2.0 — typical product - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4 - PDI: 1.90 — Experiment 3

Relevant Experimental Evidence - Example 1: ethylene-assisted metathesis of HNBR feedstock at 80° C. produced reduced molecular weight product. - Example 2: repeated ethylene metathesis at the same temperature and pressure conditions. - Example 3: lower-solids cement gave a much lower-Mooney product after metathesis. - Example 4: 1-hexene co-olefin was used and still gave a lower-MW HNBR product. - Example 1-4: metathesis reduced Mw and narrowed PDI relative to the starting polymer.

Technical Relevance This patent is highly relevant because it provides a controlled route to low-molecular-weight HNBR by combining metathesis and hydrogenation. It directly addresses the processability problem of conventional HNBR by lowering Mooney viscosity and narrowing molecular-weight distribution.

Patent 2

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the same low-molecular-weight HNBR concept as US7951875B2, using olefin metathesis of nitrile rubber followed by hydrogenation. The examples show solution processing with co-olefins and metathesis catalysts to reduce polymer chain length before hydrogenation. - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 15 wt.-% solid — Example 2 - Cement concentration: 6 wt.-% solid — Example 3 - Cement concentration: 6 wt.-% solid — Example 4 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - 1-Hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 / Examples 1-4 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - MWD: 2.0 — typical product - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4 - PDI: 1.90 — Experiment 3

Relevant Experimental Evidence - Example 1: ethylene metathesis at 80° C. reduced molecular weight. - Example 2: same process reproduced under identical ethylene conditions. - Example 3: lower-solids feed produced the lowest Mooney viscosity product. - Example 4: 1-hexene co-olefin was compatible with the metathesis reduction route. - Example 1-4: product distributions were narrower than the starting substrate.

Technical Relevance This patent is relevant because it teaches a practical route to HNBR with improved processability through pre-hydrogenation molecular-weight reduction. It complements direct hydrogenation patents by showing how metathesis can tune feedstock properties before saturation.

Patent 3

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses segment-interchange metathesis between partially hydrogenated polymers, including partially hydrogenated nitrile/butadiene rubber (HNBR), to form meta-block copolymers. The process uses a metathesis catalyst, typically Grubbs II in the examples, to couple chemically distinct polymer segments and generate ordered block-like morphologies. - Polymer mass: 0.25 g each —

Example 1 - Temperature: 95 °C — Example 1 - Catalyst amount: 8 mg Grubbs II metathesis catalyst — Example 1 - Reaction time: 1 hour — Example 1 - Polymer mass: 0.25 g each — Example 2 - Temperature: 105 °C — Example 2 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 2 - Reaction time: 1 hour — Example 2 - Polymer mass: 1.05 g fumaryl-modified polycarbonate and 0.45 g poly(ethylene-co-butadiene) — Example 3 - Temperature: 105 °C — Example 3 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 3 - Reaction time: 1 hour — Example 3 - Polymer mass: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-butadiene) — Example 4 - Temperature: 105 °C — Example 4 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 4 - Reaction time: 1 hour — Example 4 - Polymer mass: 0.25 g each — Example 5 - Temperature: 105 °C — Example 5 - Catalyst amount: 9 mg Grubbs II catalyst — Example 5 - Reaction time: 1 hour — Example 5 - Polymer mass: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) — Example 6 - Temperature: 105 °C — Example 6 - Catalyst amount: 9 mg Grubbs II metathesis catalyst — Example 6 - Reaction time: 1 hour — Example 6 - Polymer mass: 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) — Example 7 - Temperature: 105 °C — Example 7 - Catalyst loading: 8,000 ppm tungsten hexachloride — Example 7 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin — Example 7 - Reaction time: 1 hour — Example 7 - Polymer mass: 0.25 g each — Example 8 - Temperature: 105 °C — Example 8 - Catalyst amount: 9 mg Grubbs II catalyst — Example 8 - Reaction time: 1 hour — Example 8 - Polymer mass: 0.25 g SBS and 0.25 g poly(ethylene-co-butadiene) — Example 9 - Temperature: 105 °C — Example 9 - Catalyst loading: 8,000 ppm tungsten hexachloride — Example 9 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin — Example 9 - Reaction time: 1 hour — Example 9 - Polymer mass: 0.25 g fumaryl-modified polycarbonate, 0.25 g SBS, 0.25 g poly(ethylene-co-butadiene) — Example 10 - Temperature: 105 °C — Example 10 - Catalyst amount: 9 mg Grubbs II catalyst — Example 10 - Reaction time: 1 hour — Example 10 - Extraction solvent: 30 ml tetrahydrofuran — Example 5 - Extraction solvent: 30 ml tetrahydrofuran — Example 8 - Extraction solvent: 50 ml tetrahydrofuran — Example 10

Relevant Experimental Evidence - Example 1: partially hydrogenated HNBR and HPBD formed a meta-block copolymer with microphase separation. - Example 2: metathesis of unsaturated polyurethane and poly(ethylene-co-butadiene) produced segment-interchange products. - Example 7: tungsten hexachloride/tri-n-butylmethyltin enabled segment interchange in polyolefin blends. - Example 9: SBS and poly(ethylene-co-butadiene) formed an ordered meta-block copolymer. - Example 10: ternary metathesis yielded a terpolymer-like meta-block structure.

Technical Relevance This patent is relevant because it explicitly includes partially hydrogenated HNBR as a metathesis reagent, showing that hydrogenated nitrile rubber can participate in segment-interchange chemistry. It supports the broader strategy of using HNBR as a reactive polymer platform for controlled architecture and property tuning.

Patent 4

- Patent Number: US10508156B2

- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber using ruthenium- or osmium-based carbene catalysts combined with a vinyl-containing co-catalyst. The catalyst is pretreated with the co-catalyst before hydrogenation, and the examples show hydrogenation in methyl cyclohexane/bromobenzene-type solvent systems under elevated hydrogen pressure and temperature. - NBR solution concentration: 6 wt % — Example 1 - NBR mass: 18 g Perbunan® 3431VP — Example 1 - Solvent mass: 282 g MCB — Example 1 - Solvent mass: 22 g degassed MCB — Example 1 - Catalyst mass: 15 mg Wilkinson's catalyst — Example 1 - Phosphine mass: 18 mg PPh₃ — Example 1 - Hydrogenation temperature: 120° C. — Example 1 - Hydrogen pressure: 4.137 MPa — Example 1 - Agitation speed: 800 rpm — Example 1 - Hydrogenation time: 5 hours — Example 1 - Hydrogenation degree: 90.3% — Example 1 - Mn: 76,286 g/mol — Example 1 - Mw: 260,572 g/mol — Example 1 - PDI: 3.42 — Example 1 - Vinyl ethyl ether volume: 100 µL — Example 2 - Catalyst mass: 9 mg Catalyst (1) — Example 2 - Pretreatment time: 12 hours — Example 2 - Hydrogenation time: 3 hours — Example 2 - Hydrogenation degree: 93% — Example 2 - Mn: 75,844 g/mol — Example 2 - Mw: 223,863 g/mol — Example 2 - PDI: 2.95 — Example 2 - Catalyst mass: 18 mg Catalyst (2) — Example 3 - Hydrogenation time: 1 hour — Example 3 - Hydrogenation degree: 99% — Example 3 - Mn: 71,762 g/mol — Example 3 - Mw: 221,604 g/mol — Example 3 - PDI: 3.09 — Example 3 - Catalyst mass: 9 mg Catalyst (2) — Example 4 - Hydrogenation time: 2 hours — Example 4 - Hydrogenation degree: 95% — Example 4 - Mn: 71,274 g/mol — Example 4 - Mw: 208,575 g/mol — Example 4 - PDI: 2.93 — Example 4 - Catalyst mass: 9 mg Catalyst (3) — Example 5 - Hydrogenation time: 3 hours — Example 5 - Hydrogenation degree: 98% — Example 5 - Mn: 88,070 g/mol — Example 5 - Mw: 267,466 g/mol — Example 5 - PDI: 3.04 — Example 5 - NBR mass: 270 g NBR-5 — Example 6 - Solvent mass: 4,350 g MCB — Example 6 - 1-Hexene mass: 11.1 g — Example 6 - Catalyst mass: 135 mg Catalyst (2) — Example 6 - Metathesis time: 1 hour — Example 6 - Hydrogenation temperature: 140° C. — Example 6 - Hydrogen pressure: 8.4 MPa — Example 6 - Agitation speed: 600 rpm — Example 6 - Hydrogenation time: 2 hours — Example 6 - Hydrogenation degree: 99.5% — Example 6 - Mn: 4,648 g/mol — Example 6 - Mw: 9,952 g/mol — Example 6 - PDI: 2.14 — Example 6 - NBR mass: 270 g NBR-6 — Example 7 - Solvent mass: 4,350 g MCB — Example 7 - Vinyl ethyl ether mass: 0.75 g — Example 7 - Catalyst mass: 135 mg Catalyst (2) — Example 7 - Pretreatment time: 12 hours — Example 7 - Hydrogenation temperature: 140° C. — Example 7 - Hydrogen pressure: 8.4 MPa — Example 7 - Agitation speed: 600 rpm — Example 7 - Hydrogenation time: 3 hours — Example 7 - Hydrogenation degree: 99.4% — Example 7 - NBR-6 concentration: 11 wt % — Examples 8-10 - NBR mass: 560 g NBR-6 — Examples 8-10 - Solvent mass: 4,516 g MCB — Examples 8-10 - 1-Hexene mass: 23.1 g — Examples 8-10 - Catalyst

mass: 280 mg Catalyst (2) — Examples 8-10 - Metathesis time: 15 min — Examples 8-10 - VEE mass: 1.55 g — Examples 8-10 - Hydrogenation temperature: 140° C. — Examples 8-10 - Hydrogen pressure: 8.4 MPa — Examples 8-10 - Agitation speed: 600 rpm — Examples 8-10 - Hydrogenation time: 2 hours — Examples 8-10 - Hydrogenation degree: 99.8% — Examples 8-10 - Hydrogenation degree: 99.7% — Examples 8-10 - NBR-7 concentration: 13 wt % — Example 11 - NBR mass: 518 g NBR-7 — Example 11 - Catalyst mass: 363 mg Catalyst (2) — Example 11 - Hydrogenation temperature: 140° C. — Example 11 - Hydrogen pressure: 8.4 MPa — Example 11 - Agitation speed: 600 rpm — Example 11 - Hydrogenation time: 4 hours — Example 11 - Hydrogenation degree: >99% — Example 11 - Catalyst mass: 259 mg Catalyst (2) — Example 12 - Vinyl ethyl ether mass: 1.41 g — Example 12 - Hydrogenation time: 4 hours — Example 12 - Hydrogenation degree: >99% — Example 12

Relevant Experimental Evidence - Example 1: Wilkinson's catalyst with PPh₃ gave 90.3% hydrogenation in 5 hours. - Example 2: VEE pretreatment improved hydrogenation degree to 93% in 3 hours. - Example 3: Catalyst (2) with VEE reached 99% hydrogenation in 1 hour. - Example 6: metathesis plus hydrogenation produced very low-MW HNBR with PDI 2.14. - Example 12: VEE pretreatment enabled >99% hydrogenation with lower catalyst loading.

Technical Relevance This patent is directly relevant because it teaches practical hydrogenation catalyst compositions for converting NBR to HNBR with high hydrogenation degrees. It also shows that catalyst pretreatment with vinyl ethers can improve hydrogenation efficiency and help preserve molecular weight.

Patent 5

- Patent Number: US20190119411A1
- Patent Title: Process for recovery of residual hydrogenation catalyst from hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not focus on making HNBR itself, but on recovering residual hydrogenation catalyst from HNBR solution after hydrogenation. It uses a chelation step with xanthate, dithiocarbamate, or trithiocarbonate agents, followed by semi-coagulation extraction and optional washing to remove palladium and related catalyst residues. - Hydrogenated nitrile rubber content: 1% to 20% by weight — Claim 5 - Hydrogenated nitrile rubber content: 2% to 50% depending on catalyst type and mixing efficiency — Description - Hydrogenated nitrile rubber content: 2% to 13% by weight — Description - Hydrogenated nitrile rubber content: 4% to 11% by weight — Description - Reduction of original carbon-carbon double bonds: at least 80% — Description - Chelating agent amount: more than about 2 moles per 1 mole catalyst — Description - Chelation time: 15 hours — Example 2 - PEX

amount: 1.8 g — Example 2 - Acetone amount: 10 g — Example 2 - Water amount: 1.2 g — Example 2 - Semi-coagulation water: 5.9 g — Example 2 - Residual palladium: 62.1 ppm — Example 2 - Chelation time: 15 hours — Example 3 - PEX amount: 1.2 g — Example 3 - Acetone amount: 100 g — Example 3 - Water amount: 1.8 g — Example 3 - Residual palladium: 53.9 ppm — Example 3 - Chelation time: 15 hours — Example 4 - PEX amount: 1.2 g — Example 4 - Acetone amount: 96 g — Example 4 - Water amount: 1.5 g — Example 4 - Residual palladium: 35.3 ppm — Example 4 - Chelation time: 15 hours — Example 5 - PEX amount: 1.8 g — Example 5 - Acetone amount: 96 g — Example 5 - Water amount: 1.5 g — Example 5 - Residual palladium: 33.4 ppm — Example 5 - Chelation time: 17 hours — Example 6 - PEX amount: 1.8 g — Example 6 - Acetone amount: 200 g — Example 6 - Water amount: 1.5 g — Example 6 - Semi-coagulation water: 9.4 g — Example 6 - Residual palladium: 30.3 ppm — Example 6 - H-HNBR-1 cement solid content: 12.0% — Example 8 - H-HNBR-1 cement mass: 100 g — Example 8 - Residual palladium: 176.7 ppm — Example 8 - Acetone amount: 96 g — Example 8 - SDEDTC amount: 1.4 g — Example 8 - Total solid content: 6.1% — Example 8 - Chelation time: 15 hours — Example 8 - Semi-coagulation water: 5.3 g — Example 8 - Acetone wash: 90 g — Example 8 - Residual palladium: 0 ppm — Example 8 - Chelation time: 15 hours — Example 9 - SDEDTC amount: 1.8 g — Example 9 - Acetone amount: 21 g — Example 9 - Residual palladium: 86.4 ppm — Example 9 - Chelation time: 15 hours — Example 10 - SDEDTC amount: 1.2 g — Example 10 - Acetone amount: 100 g — Example 10 - Water amount: 0.6 g — Example 10 - Residual palladium: 65.4 ppm — Example 10 - Chelation time: 15 hours — Example 11 - SDEDTC amount: 1.2 g — Example 11 - Acetone amount: 96 g — Example 11 - Residual palladium: 55.4 ppm — Example 11 - Chelation time: 15 hours — Example 12 - SDEDTC amount: 1.8 g — Example 12 - Acetone amount: 96 g — Example 12 - Residual palladium: 56.1 ppm — Example 12 - Chelation time: 17 hours — Example 13 - SDEDTC amount: 1.8 g — Example 13 - Acetone amount: 235 g — Example 13 - Residual palladium: 40.9 ppm — Example 13 - Chelation time: 17 hours — Example 14 - SDEDTC amount: 0.6 g — Example 14 - Acetone amount: 129 g — Example 16 - Water amount: 1.0 g — Example 16 - PEX amount: 0.6 g — Example 16 - Chelation time: 17 hours — Example 16 - Residual palladium: 0 ppm — Example 16 - P-HNBR-1 cement solid content: 13.9% — Example 16 - P-HNBR-1 cement mass: 100 g — Example 16 - Residual palladium: 42.3 ppm — Example 16 - H-HNBR-1 cement mass: 300 g — Example 18 - Residual palladium: 176.7 ppm — Example 18 - Acetone amount: 300 g — Example 18 - SDEDTC amount: 4.3 g — Example 18 - Total solid content: 6.0% — Example 18 - Chelation time: 22 hours — Example 18 - Semi-coagulation water: 16 g — Example 18 - Residual palladium: 0 ppm — Example 18 - Chelation time: 15 hours — Example 19 - First washing time: 3 hours — Example 19 - Second washing time: 2 hours — Example 19 - Semi-coagulation water: 6 g — Example 19 - Residual palladium: 0 ppm — Example 19 - Chelation time: 10 hours — Example 20 - First washing time: 15 hours — Example 20 - Second washing time: 2 hours — Example 20 - Semi-coagulation water: 13 g — Example 20 - Residual palladium: 0 ppm — Example 20 - Chelation time: 6 hours — Example 21 - First washing time: 3 hours — Example 21 - Second washing time: 15 hours — Example 21 - Semi-coagulation water: 13 g — Example 21 - Residual palladium: 0 ppm — Example 21 - Chelation time: 15 hours — Example 22 - First washing time: 1 hour — Example 22 - Second washing time: 3 hours — Example 22 - Semi-coagulation water: 10 g — Example 22 - Residual palladium: 0 ppm — Example 22 - Chelation time: 6 hours — Example 23 - First washing time: 1 hour — Example 23 - Second washing time: 1 hour — Example 23 - Semi-coagulation

water: 10 g — Example 23 - Residual palladium: 0 ppm — Example 23

Relevant Experimental Evidence - Example 2: PEX chelation plus semi-coagulation reduced residual palladium to 62.1 ppm. - Example 6: longer chelation and more water gave 30.3 ppm residual palladium. - Example 8: chelation plus washing reduced residual palladium to 0 ppm. - Example 16: PEX-based treatment removed palladium to 0 ppm from partially hydrogenated HNBR. - Example 18-23: SDEDTC-based chelation/washing consistently achieved 0 ppm residual palladium in H-HNBR.

Technical Relevance This patent is relevant because catalyst residue removal is critical for producing clean HNBR with stable color and properties. It provides a practical post-hydrogenation purification route that complements hydrogenation patents by addressing residual palladium contamination.

Patent 6

- Patent Number: US8013215B2
- Patent Title: Production of arachidonic acid in oilseed plants
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it concerns transgenic oilseed plants producing arachidonic acid. It is included in the manifest but does not disclose hydrogenation of nitrile rubber or HNBR preparation chemistry. - Culture temperature: 26° C. — Example 1 - Shaker speed: 150 rpm — Example 1 - Light intensity: 60-85 $\mu\text{E}/\text{m}^2/\text{s}$ — Example 1 - Photoperiod: 16:8 h day/night — Example 1 - Subculture interval: every 7 days — Example 1 - Seed sterilization: 15 min in 5% Clorox solution — Example 1 - Cotyledons per plate: 25-30 — Example 1 - Incubation time: eight weeks — Example 1 - Culture temperature: 23° C. — Example 3 - Photoperiod: 16 h light, 8 h dark — Example 3 - Culture duration: 2 weeks — Example 3 - Centrifugation: 1,800 $\times$ g for 5 min — Example 3 - RNA yield: 1 mg total RNA — Example 3 - mRNA yield: 85 μg — Example 3 - PCR cycles: 35 cycles — Example 3 - PCR annealing temperature: 55° C. — Example 3 - PCR extension: 72° C. for 1 min — Example 3 - Culture temperature: 26° C. — Example 13 - Shaking speed: 150 rpm — Example 13 - Centrifugation: 1,800 $\times$ g for 10 min — Example 13 - RNA yield: 2.6 mg total RNA — Example 13 - mRNA yield: 112 μg — Example 13 - cDNA input: 224 ng mRNA — Example 13 - PCR cycles: 35 cycles — Example 13 - PCR annealing temperature: 55° C. — Example 13 - PCR extension: 72° C. for 1 min — Example 13 - PCR reaction: 94° C. for 3 min — Example 13 - PCR final extension: 72° C. for 7 min — Example 13 - ATCC deposit date: May 28, 2004 — Biological Deposits - ATCC deposit date: Jan. 30, 2003 — Biological Deposits

Relevant Experimental Evidence - Example 1: soybean embryogenic cultures and transformation workflow were described. - Example 3: *Euglena gracilis* cDNA library construction and sequencing conditions were given. - Example 13: *Pavlova lutheri* growth and cDNA synthesis conditions were detailed. - Example 15: transformed soybean embryos were analyzed for arachidonic acid production. - Example 16: construct combinations were used to generate ARA biosynthetic pathways.

Technical Relevance This patent is not technically relevant to HNBR synthesis. It does not disclose nitrile rubber hydrogenation, metathesis, or catalyst systems for HNBR preparation.

Patent 7

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses swellable rubber compositions that can include a hydrogenated nitrile elastomer as one component of a downhole swellable packer formulation. The HNBR is used as an end-use elastomer rather than being synthesized in the patent, and the disclosure emphasizes formulation, SAP loading, and swelling performance. - Curing temperature: 330 degrees Fahrenheit — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Brine concentration: 16% — Example 1 - Test temperature: 275 degrees Fahrenheit — Example 1 - Differential pressure: 5000 psi — Example 1 - SAP loading: 110 phr — Example 2 - SAP loading: 130 phr — Example 2 - SAP loading: 150 phr — Example 2 - SAP loading: 170 phr — Example 2 - Test temperature: 200 degrees Fahrenheit — Example 2 - Test temperature: 275 degrees Fahrenheit — Example 2 - Brine concentration: 5% — Example 2 - Brine concentration: 16% — Example 2 - Predicted SAP loading: about 205 Phr — Example 3 - Predicted SAP loading: about 190 Phr — Example 3 - Predicted SAP loading: about 160 Phr — Example 3 - Maximum load: about 170 Phr — Example 3 - Maximum load: about 150 Phr — Example 3 - HNBR loading: 10% — Example 4 - NBR loading: 10% — Example 4 - Swell time: 96 hour — Example 4 - Test temperature: 200° F. — Example 4 - Test temperature: 275° F. — Example 4 - Brine concentration: 5% — Example 4 - Brine concentration: 16% — Example 4 - Hydrogenation: about 70 percent to about 91 percent — Description - Mooney viscosity: about 47 to about 72 — Description - Unsaturation: about 9 percent to about 30 percent — Description - SAP loading: about 125 to about 175 Phr — Description - SAP loading: about 170 to about 210 Phr — Description - Swell: at least about 75% by volume in about 96 hours — Description - Temperature: about 200 degrees Fahrenheit — Description - Temperature: about 275 degrees Fahrenheit — Description - Pressure: at least about 3000 psi — Description - Pressure: at least about 4000 psi — Description - Pressure: at least

about 5000 psi — Description - Pressure: at least about 6000 psi — Description

Relevant Experimental Evidence - Example 1: a swellable packer cured at 330 °F and 70 psi withstood 5000 psi differential pressure. - Example 2: SAP particle size and loading were correlated with swelling and leaching behavior. - Example 3: powder SAP loading around 205 phr was predicted to achieve 75% swell in four days. - Example 4: HNBR-based packers showed improved swell rate and thermal stability versus NBR. - Example 1A-12A: claim examples define operating windows for temperature, brine, pressure, and SAP loading.

Technical Relevance This patent is relevant as an application-level disclosure showing HNBR used in a demanding swellable elastomer formulation. It highlights the importance of hydrogenation level, Mooney viscosity, and formulation balance for high-temperature, high-pressure service.

Patent 8

- Patent Number: EP3090002B1
- Patent Title: Hydrogenated nitrile rubber containing phosphine or diphosphane oxide
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenated nitrile rubber compositions containing residual phosphine or diphosphane and phosphine oxide species, produced by hydrogenation in the presence of phosphine-containing co-catalysts. The focus is on controlling residual catalyst-derived species and their effect on vulcanizate properties rather than on the hydrogenation step itself. - Phosphine content: 0 to 1.0% by weight — Claim 1 - Phosphine oxide content: 0.25 to 6% by weight — Claim 1 - Total halogen content: 25 to 10 000 ppm — Claim 1 - Hydrogenation degree: 50 to 100% — Description - Residual double bond content: maximum of about 0.9% — Description - Mooney viscosity: 10 to 120 Mooney units — Description - TPP use in hydrogenation: 5.5 wt.% — Example 3 - TPP in isolated HNBR: 0.75 wt.% — Example 3-1 - TPP = O in isolated HNBR: 1.41 wt.% — Example 3-1 - TPP in isolated HNBR: not detectable — Example 3-2 - TPP = O in isolated HNBR: 0.94 wt.% — Example 3-2 - TPP in isolated HNBR: 1.33 wt.% — Example 3-3 - TPP = O in isolated HNBR: 2.17 wt.% — Example 3-3

Relevant Experimental Evidence - Example 3-1: bromide-containing workup left measurable TPP and TPP=O in isolated HNBR. - Example 3-2: benzyl bromide led to no detectable TPP but measurable TPP=O. - Example 3-3: no reagent gave the highest TPP and TPP=O levels among the listed experiments.

Technical Relevance This patent is relevant because it addresses residual phosphine/phosphine oxide and halogen impurities in HNBR made by hydrogenation with phosphine-containing catalyst systems. Such residue control is important for downstream vulcanizate performance, especially compression set and modulus.

Patent 9

- Patent Number: US11480231B2
- Patent Title: Transmission belt and associated transmission system
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it discloses transmission belts with metallic-fiber coverings and mentions HNBR as one possible elastomeric material for belt bodies or treatments. The patent is primarily a materials/application disclosure rather than a polymer synthesis route. - Covering weight: between 500 and 1500 g/m² — Claim 1 - Metallic fibre diameter: between 2 and 40 µm — Claim 1 - Metallic fibres: at least 25% by weight of the yarn — Claim 1 - High-temperature operation: peaks of roughly 150° C. and up to around 170° C. — Description - Residual double bonds: 0.9% at most — Description - Saturation level: 4% — Description - Saturation level: 5.5% — Description - Nitrile group content: 34% — Description - Nitrile group content: 39% to 51% by weight for in-oil applications — Description - Nitrile group content: 15% to 60% — Description - Pressure: 5000 psi — Description

Relevant Experimental Evidence - No experimental examples were detected by the parser. - The description identifies HNBR as a suitable elastomer for high-temperature, in-oil belt applications. - The patent emphasizes metallic-fiber coverings rather than rubber synthesis. - HNBR is discussed in the context of residual unsaturation and saturation level requirements.

Technical Relevance This patent is only indirectly relevant to HNBR synthesis. It confirms that HNBR grades with defined saturation and nitrile content are selected for demanding belt applications, but it does not teach how to make the polymer.

Patent 10

- Patent Number: US9896424B2
- Patent Title: Methods of making sulfurized derivatives
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it discloses sulfurized derivatives of metathesis-derived functionalized monomers and polymers, mainly for lubricant and additive applications. The examples involve metathesis-derived olefinic monomers and subsequent sulfurization, not nitrile rubber hydrogenation. - 1-Decene: 1.43 moles — Example 8 - Methyl 9-decenoate: 0.179 mole — Example 9 - Di-t-butyl peroxide: 0.223 mole — Example 9 - Reaction temperature: 150° C. — Example 9 - Reaction time: 10 hr — Example 9 - 1-Decene: 1.43 moles — Example 10 - 9-Decenoic acid: 0.159 mole — Example 10 - Di-t-butyl peroxide: 0.159 mole — Example 10 - Reaction temperature: 140° C. — Example 10 - Reaction time: 3 hr — Example 10 - 1-Decene: 1.43 moles — Example 11 - 9-Decenoic acid: 0.159 mole — Example 11 - Di-t-amyl peroxide: 0.159 mole — Example 11 - Reaction temperature: 150° C. — Example 11 - Reaction time: 6.5 hr — Example 11 - Montmorillonite K10: 50 g — Example 12 - N₂ pressure: 8 psi — Example 12 - Reaction temperature: 200° C. — Example 12 - Stirring speed: 600 rpm — Example 12 - Reaction time: 8 hours — Example 12 - Vacuum distillation: 190° C. and 20 mmHg — Example 12 - 1-Decene: 1 mole — Example 13 - Methyl 9-decenoate: 1 mole — Example 13 - Montmorillonite K10: 50 g — Example 13 - N₂ pressure: 8 psi — Example 13 - Reaction temperature: 250° C. — Example 13 - Stirring speed: 600 rpm — Example 13 - Reaction time: 11 hours — Example 13 - Vacuum distillation: 190° C. and 20 mmHg — Example 13 - 1-Hexene: 1.786 mole — Example 14 - Methyl 9-decenoate: 0.179 mole — Example 14 - Reaction temperature: 150° C. — Example 14 - Reaction time: 10 hours — Example 14 - 1-Hexadecene: 1.786 mole — Example 15 - Methyl 9-decenoate: 0.179 mole — Example 15 - Reaction temperature: 150° C. — Example 15 - Reaction time: 10 hours — Example 15 - 12-Tetraeicosene: 1.786 mole — Example 16 - Methyl 9-decenoate: 0.179 mole — Example 16 - Reaction temperature: 150° C. — Example 16 - Reaction time: 10 hours — Example 16 - 1-Decene: 1.786 mole — Example 17 - Methyl 5-hexenoate: 0.179 mole — Example 17 - Reaction temperature: 130° C. — Example 17 - Reaction time: 10 hours — Example 17 - 1-Decene: 1.786 mole — Example 18 - Methyl 9-octadecenoate: 0.179 mole — Example 18 - Reaction temperature: 130° C. — Example 18 - Reaction time: 10 hours — Example 18

Relevant Experimental Evidence - Example 1: peroxide-initiated oligomerization of 1-decene/9-DAME was demonstrated. - Example 2: maleinization of 9-DAME at 200 °C gave a high-acid-value product. - Example 9: 1-decene/methyl 9-decenoate copolymerization gave a viscous product at 70% conversion. - Example 12: montmorillonite-catalyzed polymerization of methyl 9-decenoate gave an average molecular weight of about 500. - Example 18: long-chain olefin/ester copolymerization was demonstrated at 130 °C.

Technical Relevance This patent is not relevant to HNBR synthesis. It concerns sulfurized derivatives of metathesis-derived monomers for lubricant and additive applications, not hydrogenation of nitrile rubber.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7951875B2 and US7772328B2 are closely aligned: both use metathesis-first, hydrogenation-second processing to lower HNBR molecular weight and Mooney viscosity, with ethylene or 1-hexene as co-olefin and 80 °C metathesis conditions.
- US10508156B2 differs from the metathesis-first patents by focusing on direct hydrogenation of NBR at 120-140 °C and 4.137-8.4 MPa H₂, using Ru/Os-type catalyst compositions and vinyl ether pretreatment to improve hydrogenation efficiency.
- EP2057205B1 uses partially hydrogenated HNBR as a metathesis partner to build meta-block copolymers, showing that HNBR can be a reactive intermediate rather than only a final product.
- US20190119411A1 and EP3090002B1 are downstream purification/control patents: the former removes palladium residues by chelation and semi-coagulation, while the latter manages phosphine/phosphine oxide and halogen residues from hydrogenation catalyst systems.
- US10385647B2 is formulation-driven rather than synthesis-driven, but it confirms that HNBR grades with about 70-91% hydrogenation and 9-30% unsaturation are selected for swellable packers requiring high-temperature and high-pressure performance.
- US11480231B2 also is application-focused and references HNBR as a suitable elastomer for high-temperature, in-oil belt service, but it does not disclose synthesis conditions.
- US8013215B2 and US9896424B2 are outside the HNBR synthesis space; they do not teach nitrile rubber hydrogenation, though they contain general polymer or functionalization chemistry.
- Across the HNBR-relevant patents, the most repeated process variables are elevated temperature, solvent-based processing, transition-metal catalysts, and post-reaction purification or molecular-weight control to balance saturation with processability.

4. CONCLUSION

The patent evidence supports three practical synthesis directions for HNBR: direct solution hydrogenation of NBR with Ru/Os/Rh-type catalysts under elevated hydrogen pressure and temperature; metathesis-assisted molecular-weight reduction before hydrogenation to obtain lower-Mooney, narrower-distribution HNBR; and rigorous post-hydrogenation purification to remove catalyst residues and phosphine-derived impurities. For process development, the strongest evidence favors hydrogenation around 120-140 °C and 4.137-8.4 MPa H₂, with catalyst pretreatment using vinyl ethyl ether or related co-catalysts to improve activity and reduce catalyst loading. Where low viscosity is critical, metathesis with ethylene or 1-hexene at about 80 °C before hydrogenation is a validated route. For end-use quality, residual catalyst removal by chelation/semi-coagulation and control of phosphine/phosphine oxide residues are important to preserve vulcanizate performance. Overall, the evidence points to a combined strategy of

controlled hydrogenation degree, molecular-weight management, and residue cleanup as the best route to high-performance HNBR.

5. REFERENCES

- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- US10508156B2 | Catalyst compositions and their use for hydrogenation of nitrile rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10508156B2/en>
- US20190119411A1 | Process for recovery of residual hydrogenation catalyst from hydrogenated ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US20190119411A1/en>
- US8013215B2 | Production of arachidonic acid in oilseed plants | Not disclosed | US | 2011 | <https://patents.google.com/patent/US8013215B2/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- EP3090002B1 | Hydrogenated nitrile rubber containing phosphine or diphosphane oxide | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3090002B1/en>
- US11480231B2 | Transmission belt and associated transmission system | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11480231B2/en>
- US9896424B2 | Methods of making sulfurized derivatives | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9896424B2/en>